

Week 11 Workshop

Question 1

a. For Models A, B and C: $\alpha_1 + \beta_1 = 0.1 + 0.85 = 0.95$

b. Yes. Volatility persistence in a GARCH(1,1) model is measured by $\alpha_1 + \beta_1$. Since all three models have $\alpha_1 + \beta_1 = 0.95$, they have the same level of volatility persistence.

The only difference between the three models is α_0 , which affects the overall level of volatility, not the persistence.

c. See Week 9 Seminar slides 36 and 37. We also derive it in seminar in Week 10.

$$\text{var}(\varepsilon_t) = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}$$

Given that $\mu = 0$ we have $r_t = e_t$ and so

$$\text{var}(r_t) = \text{var}(\varepsilon_t) = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}$$

d. For all three models, $1 - \alpha_1 - \beta_1 = 1 - 0.1 - 0.85 = 0.05$.

Model A

$$\text{var}(r_t) = \text{var}(\varepsilon_t) = \frac{0.001}{0.05} = 0.02$$

Model B

$$\text{var}(r_t) = \text{var}(\varepsilon_t) = \frac{0.01}{0.05} = 0.20$$

Model C

$$\text{var}(r_t) = \text{var}(\varepsilon_t) = \frac{0.050}{0.05} = 1.00$$

e. Since Model C has the largest unconditional variance, it is expected to generate returns with the largest sample variance. However, in a finite simulation, the sample variance may not be exactly equal to the theoretical value.

f. Because this is a simulation question, the exact numerical sample variance depends on the random seed. A reproducible R-code is given in `Sim_garch.R`

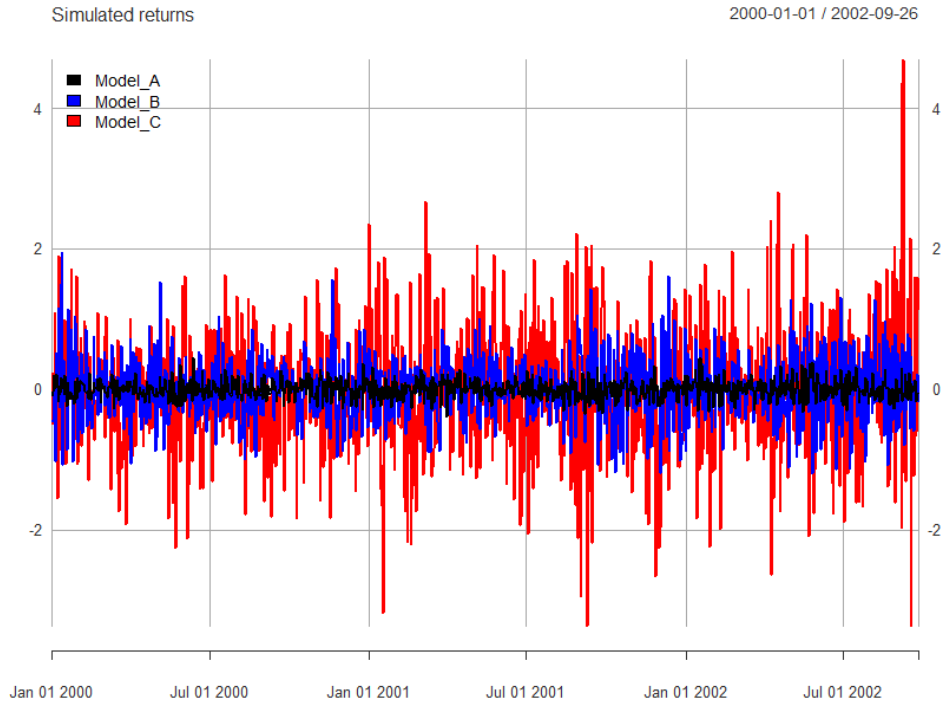
Model	Theoretical variance	Sample variance
A	0.02	0.01760105
B	0.20	0.21465410
C	1.00	0.87142012

g. The theoretical unconditional variance is the long-run average variance implied by the model. The simulated sample variances should be close to these values, especially when T is large. As T becomes very large, the sample variance should generally move closer to the theoretical unconditional variance.

h. No. Increasing α_0 increases the overall level of volatility. It increases the unconditional variance:

$$\text{var}(r_t) = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}.$$

i. No. A larger sample variance means that the returns are more dispersed overall. It does not necessarily mean that volatility is more persistent. For example, Model C has a much larger unconditional variance than Model A, but both have $\alpha_1 + \beta_1 = 0.95$. So Model C has a higher overall volatility level, but not stronger volatility clustering.



Volatility clustering refers to the tendency for high-volatility periods to be followed by high-volatility periods, and low-volatility periods to be followed by low-volatility periods. This depends mainly on $\alpha_1 + \beta_1$.

j. Model D

$$\text{var}(r_t) = \text{var}(\varepsilon_t) = \frac{0.10}{1 - (0.1 + 0.4)} = 0.20$$

Model E

$$\text{var}(r_t) = \text{var}(\varepsilon_t) = \frac{0.01}{1 - (0.1 + 0.85)} = 0.20$$

So both models have the same unconditional variance.

k. Model E has stronger volatility persistence. For Model D, $\alpha_1 + \beta_1 = 0.5$. For Model E, $\alpha_1 + \beta_1 = 0.95$. Since $0.95 > 0.5$, Model E has much stronger volatility persistence.

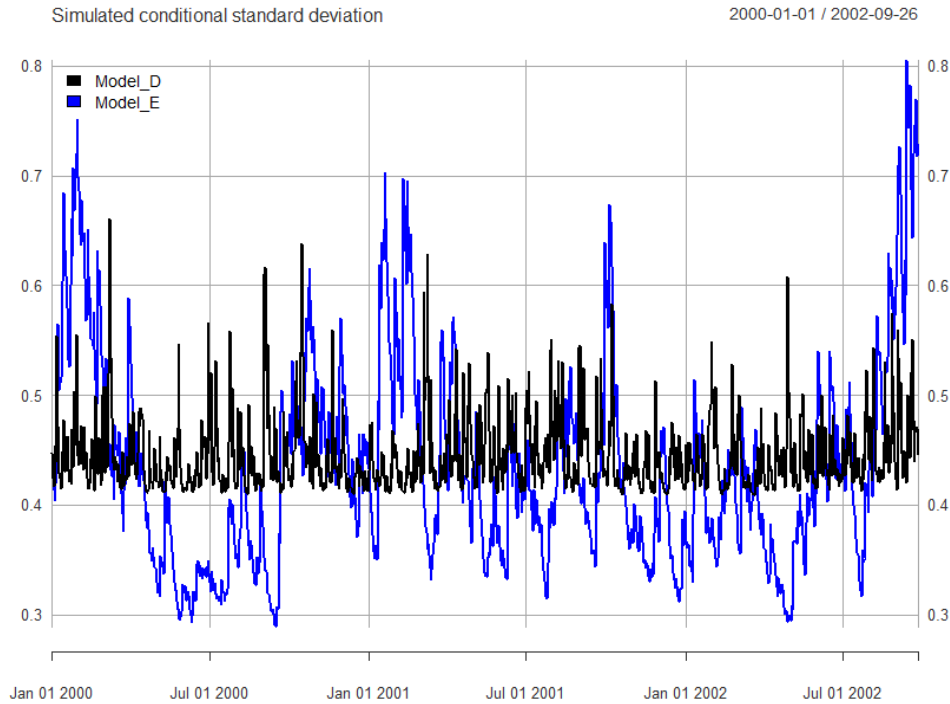
This means that shocks to volatility die out much more slowly in Model E than in Model D.

l. This example shows that two return series can have the same unconditional variance but very different volatility dynamics. Models D and E both have $\text{var}(r_t) = 0.2$. However, their

volatility persistence is very different. Model D has low persistence. Volatility shocks die out relatively quickly. Model E has high persistence. Volatility shocks die out slowly, so high-volatility periods tend to last longer.

This illustrates the usefulness of GARCH models because they allow us to distinguish between: the overall level of volatility and the persistence or clustering of volatility.

A simple constant-variance model would only focus on the average variance. A GARCH model tells us how volatility evolves over time. Therefore, even if two models have the same unconditional variance, they may imply very different risk dynamics.



In the graph, the black line (Model D) appears to fluctuate more quickly around its long-run level. When it jumps up, it tends to come back down relatively quickly. That is consistent with lower persistence. By contrast, the blue line, Model E, tends to stay high or low for longer periods. That is the stronger volatility clustering.

Question 2

(a) The estimated GARCH(1,1) model

$$\begin{aligned} r_t &= \hat{c} + \hat{\varepsilon}_t \\ \hat{\sigma}_t^2 &= \hat{\alpha}_0 + \hat{\alpha}_1 \hat{\varepsilon}_{t-1}^2 + \hat{\beta}_1 \hat{\sigma}_{t-1}^2 \end{aligned}$$

and so

$$\begin{aligned} r_t &= 0.000543 + \hat{\varepsilon}_t \\ \hat{\sigma}_t^2 &= 0.000000776 + 0.0918 \hat{\varepsilon}_{t-1}^2 + 0.9010 \hat{\sigma}_{t-1}^2 \end{aligned}$$

The 1-step ahead forecast is

$$\begin{aligned}\hat{\sigma}_{T+1}^2 &= \hat{\alpha}_0 + \hat{\alpha}_1 \hat{\varepsilon}_T^2 + \hat{\beta}_1 \hat{\sigma}_T^2 \\ &= 0.000000776 + 0.0918 (-0.003992)^2 + 0.9010 (0.0000313) \\ &= 0.0000304\end{aligned}$$

Predict a fall in the conditional variance of stock returns from 0.0000313 to 0.0000304

(b) The estimated long run variance (or unconditional variance) is

$$\widehat{var}(\varepsilon_t) = \frac{\hat{\alpha}_0}{1 - \hat{\alpha}_1 - \hat{\beta}_1} = \frac{0.000000776}{1 - 0.0918 - 0.9010} = 0.0001078$$

The fitted conditional variance $\hat{\sigma}_t^2 = 0.0000313$ on 30 September 2019 is below the long run variance.